

Artificial Intelligence

An Advanced Course About Foundations

First-Order Logic

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Extensional Semantics: tuples, relations and functions

■ Tuple

Consider a generic set of objects U

An example set of objects from U is denoted as $\{u_1, u_2\}$, where $u_1, u_2 \in U$

In a set, the order of elements is not relevant

An example of *tuple* of objects from U is denoted as $\langle u_1, u_2 \rangle$, where $u_1, u_2 \in U$

In a tuple, the order is relevant, i.e. $\langle u_1, u_2 \rangle \neq \langle u_2, u_1 \rangle$

■ Cartesian product

The cartesian product $U \times U =: U^2$ is the set of all tuples $\langle u_1, u_2 \rangle$, $u_1, u_2 \in U$

Analogously, U^3 is the set of all tuples $\langle u_1, u_2, u_3 \rangle$, $u_1, u_2, u_3 \in U$

U^4 is the set of all tuples $\langle u_1, u_2, u_3, u_4 \rangle$, $u_1, u_2, u_3, u_4 \in U$ and so on ...

■ Relation *arity is always an integer*

A relation of *arity* n is a subset of U^n

■ Function

A function of type $U^n \rightarrow U$ is a relation of arity $n + 1$ such that each tuple is constructed by associating each tuple of U^n with exactly one object from U

First-order Possible Worlds

Possible worlds made of objects, functions and relations

Each possible world is a structure $\langle U, \Sigma, v \rangle$

U is a set of object, called **domain** (also *universe of discourse*)

Σ is a set of symbols, called **signature**

v is a *function* that gives a *meaning* to the symbols in Σ with respect to U

Signature Σ

- *individual constants*: $a, b, c, d, \dots$
- *function symbols (with arity)*: $f/n, g/p, h/q, \dots$
- *predicate symbols (with arity)*: $P/k, Q/l, R/m, \dots$

Arity is an integer
that describes the expected number
of arguments

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Term

A single *individual constant* is a **term**

If f/n is a *functional symbol* (with arity n) and $t_1, \dots, t_n$ are **terms**,
then $f(t_1, \dots, t_n)$ is a **term**

Atom

If P/n is a *predicate symbol* (with arity n) and $t_1, \dots, t_n$ are **terms**,
then $P(t_1, \dots, t_n)$ is an **atom** (i.e a first-order well-formed formula – wff)

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Function v (**interpretation**)

- v assigns each *individual constant* to an *object* in U
 $v(a) \in U$ (a individual constant)
- v assigns each *functional symbol* a *function* defined on U
 $v(f/n) : U^n \rightarrow U$ (f/n functional symbol)
- v assigns each *predicate symbol* a *relation* defined on U
 $v(P/m) \subseteq U^m$ (P/m predicate symbol)

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then $f(t_1, \dots, t_n)$ is a **term**

The semantics of a **term** $f(t_1, \dots, t_n)$ is

$$v(f/n) (\langle v(t_1), \dots, v(t_n) \rangle) \in U$$

that is, the result of applying the function that v associates to f/n
to the tuple of objects in U created from the semantics of $t_1, \dots, t_n$

It is yet an object in U

First-order Language (*without variables*)

■ Well-formed formulae (wff)

All symbols in the *signature* Σ (*constants, function and predicate symbols*)

Two (primary) **logical connectives**: $\neg, \rightarrow$

Three (derived) **logical connectives**: $\wedge, \vee, \leftrightarrow$

Parenthesis: $(,)$ (there are no *precedence rules* in this language)

The definition of *terms* and *atoms* (see before)

A set of syntactic rules

The set of all the **wff** of L_{FO} is denoted as $\text{wff}(L_{FO})$

φ is an atom $\Rightarrow \varphi \in \text{wff}(L_{FO})$

$\varphi \in \text{wff}(L_{FO}) \Rightarrow (\neg\varphi) \in \text{wff}(L_{FO})$

$\varphi, \psi \in \text{wff}(L_{FO}) \Rightarrow (\varphi \rightarrow \psi) \in \text{wff}(L_{FO})$

$\varphi, \psi \in \text{wff}(L_{FO}) \Rightarrow (\varphi \vee \psi) \in \text{wff}(L_{FO}), \quad (\varphi \vee \psi) \Leftrightarrow ((\neg\varphi) \rightarrow \psi)$

$\varphi, \psi \in \text{wff}(L_{FO}) \Rightarrow (\varphi \wedge \psi) \in \text{wff}(L_{FO}), \quad (\varphi \wedge \psi) \Leftrightarrow (\neg(\varphi \rightarrow (\neg\psi)))$

$\varphi, \psi \in \text{wff}(L_{FO}) \Rightarrow (\varphi \leftrightarrow \psi) \in \text{wff}(L_{FO}), \quad (\varphi \leftrightarrow \psi) \Leftrightarrow ((\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi))$

These rules are identical to the propositional ones!

Satisfaction (*without variables*)

- Given a possible world $\langle U, \Sigma, v \rangle$

If φ is an *atom* (i.e. φ has the form $P(t_1, \dots, t_n)$)

$$\langle U, \Sigma, v \rangle \models \varphi \quad \text{iff} \quad \langle v(t_1), \dots, v(t_n) \rangle \in v(P/n)$$

If φ e ψ are wffs

$$\langle U, \Sigma, v \rangle \models (\neg \varphi) \quad \text{iff}$$

$$\langle U, \Sigma, v \rangle \not\models \varphi$$

$$\langle U, \Sigma, v \rangle \models (\varphi \rightarrow \psi) \quad \text{iff}$$

$$\text{NOT } \langle U, \Sigma, v \rangle \models \varphi \quad \text{OR} \quad \langle U, \Sigma, v \rangle \models \psi$$

$$\langle U, \Sigma, v \rangle \models (\varphi \wedge \psi) \quad \text{iff}$$

$$\langle U, \Sigma, v \rangle \models \varphi \quad \text{AND} \quad \langle U, \Sigma, v \rangle [s] \models \psi$$

$$\langle U, \Sigma, v \rangle \models (\varphi \vee \psi) \quad \text{iff}$$

$$\langle U, \Sigma, v \rangle \models \varphi \quad \text{OR} \quad \langle U, \Sigma, v \rangle [s] \models \psi$$

What is *true*?

- **A world of cats**

Likes	Tom	Spot	Kitty	Felix
Tom	x			
Spot	x		x	
Kitty		x	x	
Felix			x	

translates into $\langle U, \Sigma, v \rangle$

- **Universe**

$U := \{\underline{\text{tom}}, \underline{\text{spot}}, \underline{\text{kitty}}, \underline{\text{felix}}\}$ ← *Could not put real cats in U :)
underlined names here stand for objects*

- **Signature**

$\Sigma := \{\text{tom}, \text{spot}, \text{kitty}, \text{felix}, \text{Likes}/2\}$ *four constants and one predicate symbol*

- **Interpretation**

$v(\text{tom}) = \underline{\text{tom}}, \quad v(\text{spot}) = \underline{\text{spot}}, \quad v(\text{kitty}) = \underline{\text{kitty}}, \quad v(\text{felix}) = \underline{\text{felix}},$

$v(\text{Likes}/2) =$ *a subset of $U \times U$*

$\{\langle \underline{\text{tom}}, \underline{\text{tom}} \rangle, \langle \underline{\text{spot}}, \underline{\text{tom}} \rangle, \langle \underline{\text{spot}}, \underline{\text{kitty}} \rangle, \langle \underline{\text{kitty}}, \underline{\text{spot}} \rangle, \langle \underline{\text{kitty}}, \underline{\text{kitty}} \rangle, \langle \underline{\text{felix}}, \underline{\text{kitty}} \rangle\}$

What is *true*?

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■ Sentences

$\langle U, \Sigma, v \rangle \models Likes(spot, kitty)$ because $\langle v(spot), v(kitty) \rangle \in v(Likes/2)$

$\langle U, \Sigma, v \rangle \models Likes(tom, tom)$ because $\langle v(tom), v(tom) \rangle \in v(Likes/2)$

$\langle U, \Sigma, v \rangle \models \neg Likes(kitty, felix)$ because $\langle v(kitty), v(felix) \rangle \notin v(Likes/2)$

$\langle U, \Sigma, v \rangle \not\models Likes(tom, kitty)$ because $\langle v(tom), v(kitty) \rangle \notin v(Likes/2)$

$\langle U, \Sigma, v \rangle \not\models \neg Likes(felix, kitty)$ because $\langle v(felix), v(kitty) \rangle \in v(Likes/2)$

What is *true*?

■ A world of cats

Likes	Tom	Spot	Kitty	Felix
Tom	x			
Spot	x		x	
Kitty		x	x	
Felix			x	

translates into $\langle \mathbf{U}, \Sigma, v \rangle$

■ Sentences

$$\langle \mathbf{U}, \Sigma, v \rangle \models (\text{Likes}(\text{spot}, \text{kitty}) \wedge \text{Likes}(\text{felix}, \text{kitty}))$$

$$\langle \mathbf{U}, \Sigma, v \rangle \models (\text{Likes}(\text{tom}, \text{kitty}) \vee \text{Likes}(\text{tom}, \text{tom}))$$

$$\langle \mathbf{U}, \Sigma, v \rangle \models (\text{Likes}(\text{spot}, \text{tom}) \vee \neg \text{Likes}(\text{spot}, \text{tom}))$$

is satisfied in this possible world but also in any possible world

Variables & Quantifiers

■ Well-formed formulae (wff)

All symbols in the *signature* Σ (i.e. *constants, function and predicate symbols*)

A set of **variables**: x, y, z

Two (primary) **logical connectives**: $\neg, \rightarrow$

Three (derived) **logical connectives**: $\wedge, \vee, \leftrightarrow$

Two **quantifiers**: $\forall, \exists$

Parentheses: $(,)$ (there are no *precedence rules* in this language)

An extended definition of *terms* and *atoms*

Term

A single *individual constant* or a **variable** is a **term**

If f/n is a *functional symbol* (with arity n) and $t_1, \dots, t_n$ are **terms**,
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An extended definition of *terms* and *atoms* (see before)

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$\varphi \in \text{wff}(L_{FO}) \Rightarrow (\forall x \varphi) \in \text{wff}(L_{FO})$

$\varphi \in \text{wff}(L_{FO}) \Rightarrow (\exists x \varphi) \in \text{wff}(L_{FO})$

 x can be any variable

Satisfaction

- Given a possible world $\langle U, \Sigma, v \rangle$ and a valuation s (on that world)

A valuation is a function $s : \text{Variables} \rightarrow U$

← In a sense, a valuation s transforms all variables into constants

If φ is an atom (i.e. φ has the form $P(t_1, \dots, t_n)$)

$\langle U, \Sigma, v \rangle [s] \models \varphi$ iff $\langle v(t_1) [s], \dots, v(t_n) [s] \rangle \in v(P) [s]$

If φ e ψ are wffs

$\langle U, \Sigma, v \rangle [s] \models (\neg \varphi)$ iff

$\langle U, \Sigma, v \rangle [s] \not\models \varphi$

$\langle U, \Sigma, v \rangle [s] \models (\varphi \wedge \psi)$ iff

$\langle U, \Sigma, v \rangle [s] \models \varphi$ AND $\langle U, \Sigma, v \rangle [s] \models \psi$

$\langle U, \Sigma, v \rangle [s] \models (\varphi \vee \psi)$ iff

$\langle U, \Sigma, v \rangle [s] \models \varphi$ OR $\langle U, \Sigma, v \rangle [s] \models \psi$

$\langle U, \Sigma, v \rangle [s] \models (\varphi \rightarrow \psi)$ iff

NOT $\langle U, \Sigma, v \rangle [s] \models \varphi$ OR $\langle U, \Sigma, v \rangle [s] \models \psi$

Quantified formulae

$\langle U, \Sigma, v \rangle [s] \models \forall x \varphi$ iff

FORALL $\underline{d} \in U$ we have $\langle U, \Sigma, v \rangle [s](x:\underline{d}) \models \varphi$

$\langle U, \Sigma, v \rangle [s] \models \exists x \varphi$ iff

it EXISTS $\underline{d} \in U$ such that $\langle U, \Sigma, v \rangle [s](x:\underline{d}) \models \varphi$

Where $[s](x:\underline{d})$ is the *variant* of function s that assigns $\underline{d}$ to x and remains unaltered for any other variables.

What is *true*?

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Tom	x			
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translates into $\langle U, \Sigma, v \rangle$

- Sentences

$\langle U, \Sigma, v \rangle [s] \models (\forall x (\exists y Likes(x, y)))$ because

FORALL cat1 $\in U$, $\langle U, \Sigma, v \rangle [s](x:\text{cat1}) \models (\exists y Likes(x, y))$ because

it EXISTS cat2 $\in U$, $\langle U, \Sigma, v \rangle ([s](x:\text{cat1}))(y:\text{cat2}) \models Likes(x, y)$

What is *true*?

- A world of cats

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Tom	x			
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Felix			x	

translates into $\langle U, \Sigma, v \rangle$

- Sentences

$\langle U, \Sigma, v \rangle [s] \not\models (\exists x (\forall y Likes(x, y)))$ because

it EXISTS cat1 $\in U$, $\langle U, \Sigma, v \rangle [s](x:\text{cat1}) \not\models (\forall y Likes(x, y))$ because

NOT FORALL cat2 $\in U$, $\langle U, \Sigma, v \rangle ([s](x:\text{cat1}))(y:\text{cat2}) \models Likes(x, y)$

Variables & Quantifiers: further examples

- “Being brothers means being relatives”

$$\forall x \forall y (Brother(x, y) \rightarrow Relative(x, y))$$

- “Being relative is a symmetric relation”

$$\forall x \forall y (Relative(x, y) \leftrightarrow Relative(y, x))$$

- “By definition, being mother is being parent and female”

$$\forall x (Mother(x) \leftrightarrow (\exists y Parent(x, y) \wedge Female(x)))$$

- “A cousin is a son of either a brother or a sister of either parents”

$$\begin{aligned} \forall x \forall y (Cousin(x, y) \\ \leftrightarrow \exists z \exists w (Parent(z, x) \wedge Parent(w, y) \wedge (Brother(z, w) \vee Sister(z, w)))) \end{aligned}$$

- “Everyone has a mother”

$$\forall x \exists y Mother(y, x)$$

BE CAREFUL about the ordering of quantifiers, in fact:

$$\exists y \forall x Mother(y, x)$$

“There is one (common) mother to everyone”

Open formulae, Sentences

- **Bound** and **free** variables

The occurrence of a *variable* in a wff is **bound** if it lies in the scope of a quantifier (for that *variable*)

The occurrence of a *variable* in a wff is **free** if it is not *bound*

Examples of bound variables:

$$\forall x P(x)$$

$$\exists x (P(x) \rightarrow (A(x) \wedge B(x)))$$

Examples of free variables:

$$P(x)$$

$$\exists y (P(y) \rightarrow (A(x,y) \wedge B(y)))$$

- **Open** and **closed** formulae: **sentences**

A wff is **open** if there is at least one free occurrence of a variable

Otherwise, the wff is **closed** (also called **sentence**)

Only *closed* wffs, i.e. *sentences*, have a truth value (see after)

Models

- **Validity** in a possible world, *model*

A wff φ such that $\langle U, \Sigma, v \rangle [s] \models \varphi$ for any valuation s is **valid** in $\langle U, \Sigma, v \rangle$

$\langle U, \Sigma, v \rangle$ is also a *model* of φ

$\langle U, \Sigma, v \rangle \models \varphi$ (i.e. the reference to s can be omitted)

A possible world $\langle U, \Sigma, v \rangle$ is a *model* of a set of wff Γ iff it is a model for all the wffs in Γ

$\langle U, \Sigma, v \rangle \models \Gamma$

- **Truth**

A **sentence** ψ such that $\langle U, \Sigma, v \rangle [s] \models \psi$ for one valuation s is **valid** in $\langle U, \Sigma, v \rangle$

If the sentence is true for one valuation s , then is true for all valuations

A **sentence** ψ is **true** in $\langle U, \Sigma, v \rangle$ if it is **valid** in $\langle U, \Sigma, v \rangle$

Validity in general

■ Validity and logical truth

A wff (either open or closed) is **valid** (also **logically valid**) if it is **valid** in any possible world $\langle U, \Sigma, v \rangle$

Example:

$$(P(x) \vee \neg P(x))$$

A sentence ψ is a **logical truth** if it is **true** in any possible world $\langle U, \Sigma, v \rangle$

we write then $\models \psi$ (i.e. no reference to $\langle U, \Sigma, v \rangle$)

Examples:

$$\forall x (P(x) \vee \neg P(x))$$

$$\forall x \forall y (G(x,y) \rightarrow (H(x,y) \rightarrow G(x,y)))$$

■ Inconsistence

A wff (either open or closed) is **inconsistent** if its not *satisfiable*

Example:

$$\forall x (P(x) \wedge \neg P(x))$$

Entailment

- Definition

Given a set of wffs Γ and one wff φ , we have

$$\Gamma \models \varphi$$

iff all possible worlds $\langle U, \Sigma, v \rangle [s]$ satisfying Γ also satisfy φ

This definition embraces *all* possible combinations $\langle U, \Sigma, v \rangle [s]$

The only thing that does not vary is the language Σ

Is this problem **decidable**?

In general, in first-order logic, a direct calculus of entailment is impossible...